

Seat No:

SET	P
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AUTONOMOUS STATE EXAMINATION BOARD**B.Tech Semester VI: Design & Analysis of Algorithms****Subject: Design & Analysis of Algorithms (Theory & Proofs) · Pattern: CBCS Pattern****Day & Date:** Thursday, September 17, 2026 **Duration:** 180 Minutes **Max. Marks:** 70 Marks**Instructions for Candidates:**

1. Q.1 is compulsory. Mention the question paper set (**P**) clearly on top of the answer booklet.
2. Figures to the right indicate full marks assigned to each question.
3. Assume suitable data wherever necessary and state your assumptions explicitly.
4. Use of programmable calculators or unauthorized electronic communication devices is strictly prohibited.
5. Standard University Marking Scheme

Q.1 Choose the correct alternatives for the following questions. [14 Marks]

1. Digitalizing a picture definition into a set of intensity values is known as _____.
a) Scanning b) Refreshing
b) Scan conversion d) All
2. _____ is the ratio of horizontal points to vertical points necessary to produce equal length lines in both directions.
a) Aspect ratio b) Resolution
b) Dot pitch d) Height weight ratio
3. In Circle algorithm only one octant of the circle is generated and the other parts are obtained by successive _____.
a) Rotation b) Reflections
b) Refraction d) Scaling
4. _____ touchscreen technology detects your fingers using sound instead of light.
a) Infrared b) Optical
b) SAW d) Capacitive
5. The process of mapping a world window in world co-ordinate system to view port are called as _____.
a) View port b) Transformation viewing
b) Clipping window d) All
6. In homogeneous coordinates, lower right hand 1×1 submatrix in 3D transformations produces the _____.

- a) Overall scaling b) Perspective transformation
b) Projective transformation d) Translation
7. Figures to the right indicate full marks. Section I Q.2 Solve any three 12 Q.3 Solve any two 16 Write a short note on Interactive devices. i) Joysticks ii) Light pens st quadrant. Combined transformation on the object. i) Scale the object 2 units in X-direction and 2 units in Y-direction. ii) Translate the object 3 units in X-direction and 5 units in Y-direction. iii) Rotate the object with the angle of 270 degree. Section II Q.4 Solve any three 12 Q.5 Solve any two 16 Subdivision algorithm? data structure.
- a) Elaborate any 4 applications of computer Graphics.
b) Differentiate between random scan display and raster scan display.
c) Elaborate edge fill algorithm.
d) Homogeneous Coordinate system in 3D Transformation.
e) a)
f) Consider the origin centered circle with radius 5 and generate circle for 1
g) Consider the 2D object with the vertices A(1,3) B(3,6) C(3,3) -Apply
h) Elaborate working of Cohen-Sutherland line clipping algorithm.
i) Explain Painters algorithm.
j) Elaborate various properties of Bezier curves.
k) Differentiate Non parametric & parametric curves.
l) How can you identify visible and invisible line segments using Mid-point
m) Illustrate sub-division process in Warnock algorithm and draw quad tree
n) Elaborate B-Spline curves and its blending function.
8. The algorithm divides 2D space into 9 regions, of which only the middle part (view port) is known as -----
a) Cohen-Sutherland b) Liang Barsky
b) Sutherland-Hodgeman d) N L N
9. The Bezier Curve always passes through ----- control point.
a) First b) Last
b) Both first and last d) All
10. According to Cohen-Sutherland algorithm a line is completely outside the window if -----
a) The region code of line end points have a 1 in same bit position
b) The end points region code is non-zero values
c) If L bit and R bit are non-zero
d) The region code of line end points have a 0 in same bit position
11. ----- display uses the primary gun and flood gun to store the picture pattern and to maintains the picture display respectively.
a) Direct View Storage Tube b) Beam penetration
b) Shadow masking d) Thin-film electroluminescent

12. Co-ordinates of window are known as -----
- a) Screen co-ordinates
 - b) World co-ordinates
 - c) Device co-ordinates
 - d) Cartesian co-ordinates
13. The rectangle portion of interface window that defines where image will actually appear are called -----
- a) Transformation
 - b) View port
 - c) Clipping window
 - d) Screen co-ordinates
14. Circle algorithm begins at ____ for counter clock wise generation of the circle.
- a) $(y = 0, x = R)$
 - b) $(y = 0, x = 0)$
 - c) $(y = R, x = R)$
 - d) $(y = R, x = 0)$
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— END OF QUESTION PAPER (P) —